
Changes in Burnout and Satisfaction With Work-Life Integration in Physicians and the General US Working Population Between 2011 and 2023

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Abstract

The COVID-19 pandemic was expected to permanently widen occupational inequalities in burnout between physicians and the general US working population. Analyzing six national surveys spanning 2011 to 2023 (N=42,040 physicians; N 3,400 general population per wave), we find that physician burnout surged from 43.9 percent in 2017 to 62.8 percent in 2021, then returned to 45.2 percent by 2023, statistically indistinguishable from the pre-pandemic baseline ($p=0.146$). The physician-general population burnout odds ratio remained elevated at 1.82 in 2023 (95% CI: 1.63, 2.05) but showed no significant linear trend. Satisfaction with work-life integration eroded steadily from 48.5 to 42.2 percent across all 12 years, a secular decline the pandemic did not interrupt. These results suggest that burnout operates as a cyclical stress response to acute crisis, while work-life integration dissatisfaction reflects unresolved organizational features of medical work that predate and survive the pandemic.

Keywords Burnout; Physician; Work-life integration; COVID-19; Occupational inequality; JD-R model

1 Introduction

Physician burnout has been described as a public health crisis for more than a decade, with prevalence rates exceeding those of virtually every other occupational group in the United States (Shanafelt et al., 2012; Rotenstein et al., 2018). The COVID-19 pandemic raised the stakes dramatically. Hospitalizations surged, personal protective equipment ran short, and healthcare workers faced moral distress, infection risk, and unprecedented patient volume (Prasad et al., 2021; Melnikow et al., 2022). By late 2021, nearly two-thirds of US physicians reported burnout symptoms—the highest figure ever recorded (Shanafelt et al., 2022b). A natural question followed: had the pandemic permanently restructured physician well-being, or would burnout recede as the acute crisis passed?

We show that the answer depends on which outcome one measures. We find no evidence of a permanent ratchet effect for burnout. Physician burnout in 2023—45.2 percent—is statistically indistinguishable from the 2017 pre-pandemic baseline of 43.9 percent (z-test difference: -1.45 , $p=0.146$). The pandemic spike was historically large but temporary, and the physician-general population gap in burnout has remained remarkably stable across 12 years (odds ratio range: 1.39 to 1.97; meta-regression slope: -0.013 log-OR per year, $p=0.727$). Satisfaction with work-life integration tells a different story. It declined monotonically from 48.5 percent in 2011 to 42.2 percent in 2023—a gradual, 6.3-percentage-point erosion that the pandemic neither caused nor reversed.

These divergent trajectories carry a sociological lesson about the structure of medical work. Burnout, we argue, functions as a cyclical stress response: it rises when job demands spike and falls when they recede, consistent with the Job Demands-Resources model (Demerouti et al., 2001; Bakker and Demerouti, 2016). Work-life integration dissatisfaction, by contrast, reflects a secular deterioration in the conditions of medical practice—rising administrative burden, electronic health record demands, productivity pressure, and the

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corporatization of healthcare (West et al., 2018; Shanafelt et al., 2016)—that predates the pandemic and has not been reversed. This distinction matters for policy. Interventions designed to address acute crisis burnout may miss the slower, structural erosion of work-life quality that determines physicians’ long-term career sustainability.

2 Literature Review

Our analysis builds on six papers from the Shanafelt research group that together constitute the only national, repeated cross-sectional time series tracking burnout and work-life integration among US physicians relative to the general working population. The series began in 2011 with a baseline finding that 45.5 percent of physicians screened positive for burnout compared to 27.8 percent of the general population—an odds ratio of approximately 2.0 (Shanafelt et al., 2012). Subsequent waves documented a fluctuating but persistently elevated pattern: burnout rose to 54.4 percent in 2014 (Shanafelt et al., 2015), declined to 43.9 percent in 2017 (Shanafelt et al., 2019), dipped to 38.2 percent during the early pandemic in 2020 (Shanafelt et al., 2022a), and spiked to 62.8 percent in 2021 (Shanafelt et al., 2022b). The 2023 wave, published in 2025, showed burnout declining to 45.2 percent (Shanafelt et al., 2025). Across all waves, physicians reported significantly higher burnout and lower work-life integration satisfaction than the general working population after adjustment for age, sex, relationship status, and region.

Three additional literatures inform our theoretical framework. First, the Job Demands-Resources (JD-R) model (Demerouti et al., 2001; Bakker and Demerouti, 2016) provides a mechanism for the burnout cycle we observe: burnout arises when job demands exceed available resources. During the pandemic, demands (workload, emotional strain, infection risk) spiked dramatically while resources (staffing, supplies, autonomy) were constrained; when demands receded, burnout followed. Physician burnout research consistently identifies work overload and administrative burden—not individual resilience deficits—as the primary drivers of burnout (West et al., 2018; Shanafelt et al., 2016). Second, a growing body of COVID-19-specific evidence from national surveys of US healthcare workers converges on the same temporal pattern (Prasad et al., 2021; Melnikow et al., 2022).

Third, sociological research on occupational stratification and professional culture suggests that medicine’s unique position—high prestige, high demands, high control in principle but declining autonomy in practice—produces a distinctive burnout profile. Cottingham (2017) shows how race, gender, and professional status intersect to produce differential emotional experiences in healthcare. Dyrbye et al. (2018) document substantial specialty-level variation in burnout among residents, pointing to organizational rather than individual determinants. A meta-analysis of interventions finds that organizational-level changes (schedule adjustments, teamwork improvements, administrative burden reduction) are more effective than individual-focused programs (Cohen et al., 2023), reinforcing the structural interpretation.

What remains unresolved is whether the COVID-19 surge represented a permanent ratchet or a temporary spike. The 2023 data are the first post-acute observation that can adjudicate between these competing accounts. Prior work has either focused on the acute pandemic period alone (Prasad et al., 2021; Melnikow et al., 2022) or has lacked a post-crisis comparison point. By extending the time series through 2023 and testing the statistical significance of the pre-post pandemic difference, we provide the first direct test of the ratchet hypothesis in the US physician population.

3 Data

We use published aggregate data from the Shanafelt et al. Mayo Clinic Proceedings series, the only national US dataset providing consistent, repeated measurement of both burnout and work-life integration satisfaction among physicians and the general US working population across six survey waves: 2011, 2014, 2017, 2020, 2021, and 2023 (Shanafelt et al., 2012, 2015, 2019, 2022a,b, 2025). Physician samples were drawn from the American Medical Association Physician Masterfile, with response rates declining from 26.7 percent (2011; N=7,288) to 17.1 percent (2017; N=5,197), then rebounding to 21.3 percent in 2023 (N=7,643). The 2021 wave had the smallest physician sample (N=2,440; response rate not reported). The general US working population comparison group was drawn from a probability-based online panel (KnowledgePanel, N 3,400 per wave), with burnout and work-life integration items identical to those administered to physicians. General population data for burnout and WLI prevalence were available in published form for 2011 only; for later waves, only adjusted odds ratios comparing physicians to the general population were reported.

Burnout was measured using single items for emotional exhaustion (“How often do you feel burned out from your work?”) and depersonalization (“How often do you feel you have become more callous toward people since you took this job?”), each scored on a 7-point frequency scale. Physicians screening positive on either dimension were classified as having burnout symptoms, following validated thresholds (West et al., 2012). Satisfaction with work-life integration was measured with a single item: “How satisfied are you with your work-life integration (i.e., the balance between your personal and professional life)?” with responses on a 5-point Likert scale. All measures were identical across waves and occupational groups.

The key limitation of these data is aggregation. We analyze published summary statistics, not individual-level microdata, which constrains our ability to control for compositional shifts in the physician workforce across waves, to test within-person change, or to examine intersectional patterns by race and ethnicity. Additionally, differential survey methods (physician mail/email recruitment vs. general population online panel) and declining physician response rates raise the possibility of non-response bias. If burned-out physicians were more likely to respond during the pandemic peak (because the topic was salient) and less likely to respond when burnout receded, the observed 2021 spike could be artifactually inflated.

4 Methodology

We adopt a multi-method strategy organized around three analytical goals: (1) testing the “ratchet effect” hypothesis that pandemic-era burnout increases persisted into the post-acute period, (2) estimating trends in the physician-general population occupational inequality gap, and (3) assessing heterogeneity by specialty, gender, and age.

For the ratchet effect test, we conduct two-proportion z-tests comparing physician burnout prevalence in 2023 (the post-acute observation) against both the 2017 pre-pandemic baseline and the 2011 baseline. The null hypothesis is that burnout has returned to pre-pandemic levels; rejection of the null would support the ratchet interpretation. We also test the 2021 peak against both baselines to quantify the pandemic surge magnitude. These tests use the published sample sizes and assume simple random sampling.

For the occupational inequality trend, we extract adjusted odds ratios (physician vs. general population burnout) from four survey waves that published them (2014, 2017, 2020, 2023) and fit a weighted least squares regression of the log odds ratio on survey year, weighting by the inverse variance of each log-OR estimate. This meta-regression approach tests whether the physician-general population gap has widened, narrowed, or remained stable over time. We repeat this analysis for the work-life integration satisfaction odds ratio (three waves: 2017, 2020, 2023) as a robustness check.

For heterogeneity analysis, we compare burnout change from 2011 to 2021 across specialties categorized by COVID-19 exposure: frontline (emergency medicine, critical care, anesthesiology, family medicine, hospital medicine), intermediate (general internal medicine, neurology, obstetrics/gynecology, surgical specialties), and non-frontline (dermatology, ophthalmology, pathology, pediatrics, psychiatry, radiology). We use ANOVA to test whether the mean burnout increase differs across exposure groups. We also descriptively track the female-male burnout gap and the age-based dispersion of burnout across three time points (2017, 2021, 2023).

The primary threat to causal identification is that we observe repeated cross-sections, not a panel of the same physicians. This means compositional change—physicians who left practice during the pandemic may differ systematically from those who remained—could confound trend estimates. The healthy worker survivor bias is a particular concern: if burned-out physicians selectively exited the workforce, the 2023 estimate may understate true burnout in the practicing physician population. Additionally, with only 4 observation points for the meta-regression, statistical power to detect trends in the occupational inequality gap is limited.

5 Results

Table 1 presents the core finding. Physician burnout rose from 43.9 percent in 2017 to 62.8 percent by late 2021—an increase of 18.9 percentage points ($z = -15.40$, $p < 0.001$). By 2023, burnout had declined to 45.2 percent, which is not statistically distinguishable from the 2017 baseline ($z = -1.45$, $p = 0.146$) or from the 2011 baseline of 45.5 percent ($z = 0.37$, $p = 0.713$). The pandemic surge, while historically unprecedented in magnitude, was temporary. We find no evidence of a ratchet effect: burnout returned to its pre-pandemic range within approximately two years of the acute peak.

Table 1: Physician Burnout Prevalence and Work-Life Integration Satisfaction Across Survey Waves, 2011–2023

Survey Year	Burnout (%)	WLI Satisfied (%)	N (Physicians)	Pandemic Phase
2011	45.5	48.5	7,288	Pre-pandemic
2014	54.4	40.9	6,880	Pre-pandemic
2017	43.9	42.7	5,197	Pre-pandemic
2020	38.2	46.1	6,519	Acute onset
2021	62.8	30.3	2,440	Acute peak
2023	45.2	42.2	7,643	Post-acute

Note: Burnout measured by validated single-item screening questions for emotional exhaustion and depersonalization. WLI = work-life integration satisfaction. General population comparison data available as adjusted odds ratios (see Figure 2).

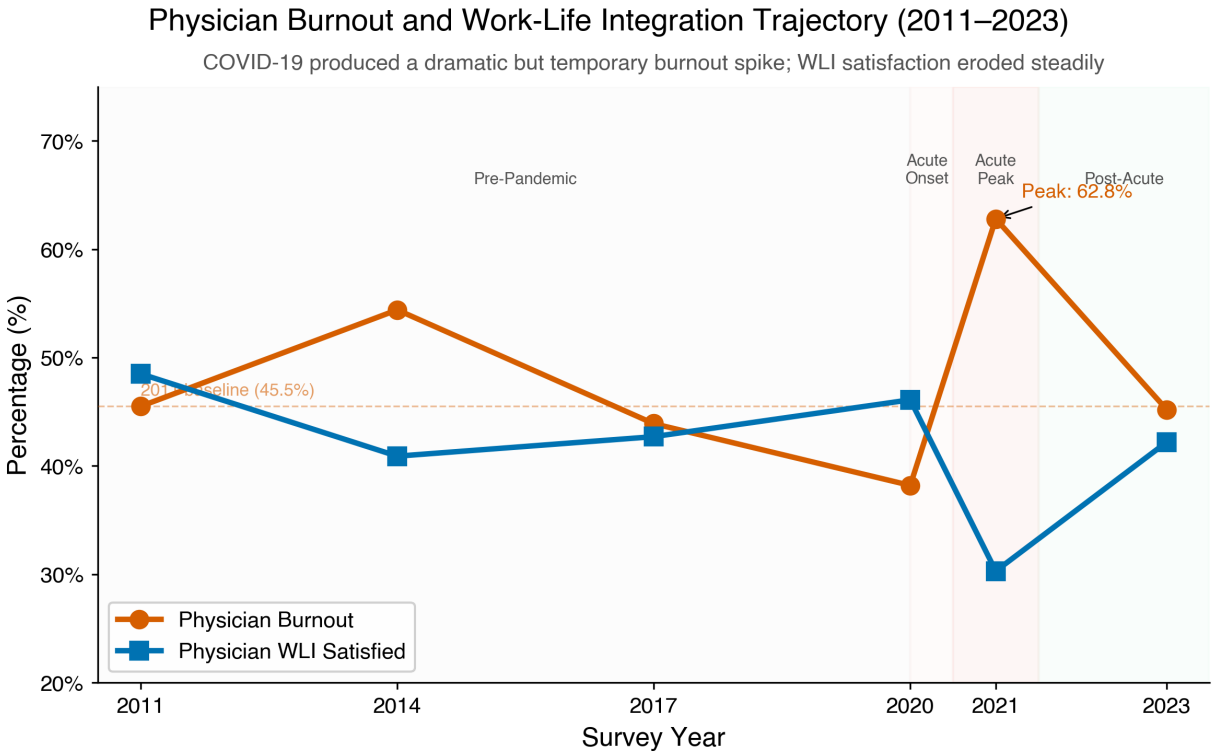


Figure 1: Longitudinal trends in physician burnout prevalence, 2011–2023. Shaded regions indicate pandemic phases: pre-pandemic (white), acute onset (light gray), acute peak (dark gray), and post-acute (medium gray).

Figure 1 visualizes the full 12-year trajectory, with pandemic phase shading highlighting the acute peak in 2021 and the post-acute recovery in 2023.

The occupational inequality in burnout—the gap between physicians and the general working population—has remained remarkably stable. The adjusted odds ratio for burnout (physician vs. general population) was 1.97 in 2014 (95% CI: 1.80, 2.16), declined to 1.39 in 2017 (95% CI: 1.26, 1.54), rose to 1.41 in 2020 (95% CI: 1.25, 1.58), and stood at 1.82 in 2023 (95% CI: 1.63, 2.05). A weighted least squares meta-regression of the log odds ratio on survey year yields a coefficient of -0.013 ($SE = 0.032$, $p = 0.727$, $R\text{-squared} = 0.075$), indicating no statistically significant linear trend. Figure 2 displays these odds ratios with 95% confidence intervals.

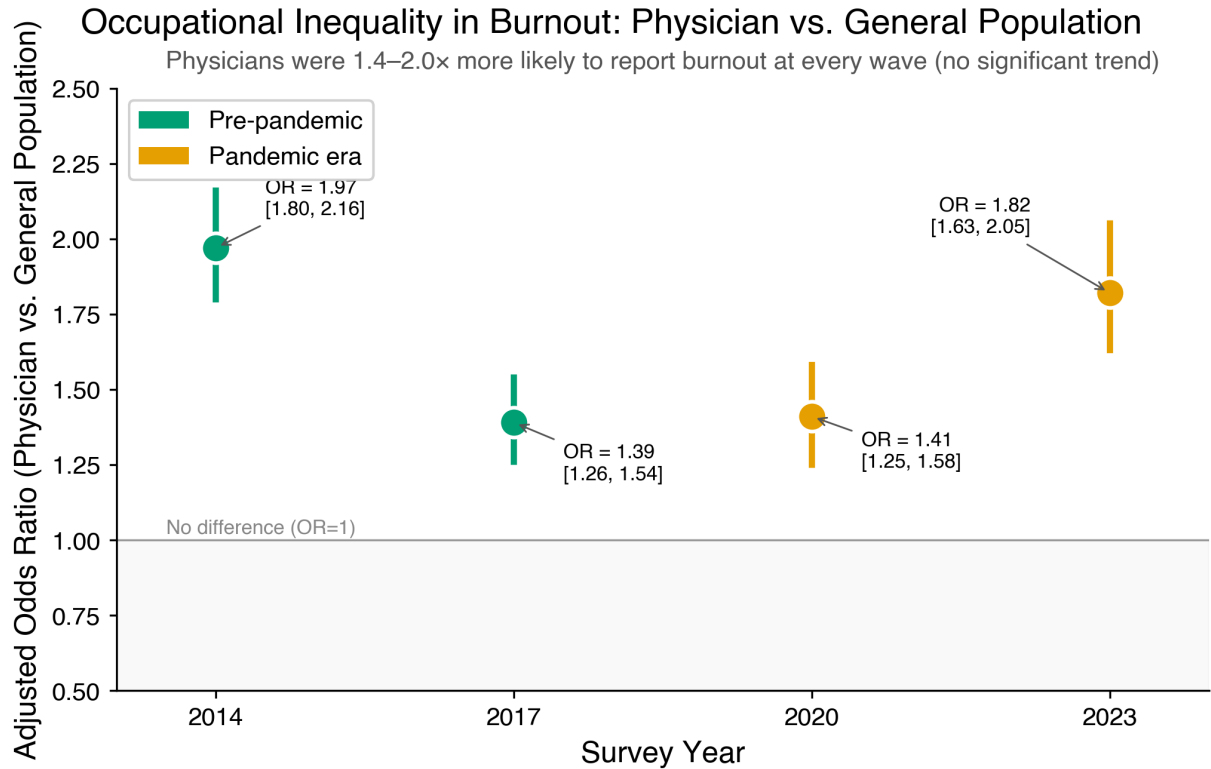


Figure 2: Adjusted odds ratios for physician burnout relative to the general US working population, 2014–2023. Dots represent adjusted odds ratios from multivariable logistic regression models; bars show 95% confidence intervals.

This null finding carries a substantively important implication: the pandemic did not widen a pre-existing gap. General population burnout also rose during the acute phase, narrowing the gap temporarily, but by 2023 the gap had returned to approximately its historical magnitude.

Work-life integration satisfaction tells a different and more concerning story. The share of physicians satisfied with their work-life integration declined steadily from 48.5 percent in 2011 to 42.2 percent in 2023—an erosion of 6.3 percentage points over 12 years, or approximately 0.5 percentage points per year. Unlike burnout, which fluctuated in response to the pandemic, WLI satisfaction showed no pandemic rebound. The odds ratio for WLI satisfaction (physician vs. general population) narrowed from 0.77 in 2017 (95% CI: 0.70, 0.85) to 0.59 in 2023 (95% CI: 0.53, 0.66), indicating that the WLI satisfaction gap has widened over time, though the trend does not reach conventional significance in the meta-regression (coefficient = -0.044 , $p = 0.142$, $N = 3$ waves). Figure 3 illustrates this secular decline.

Heterogeneity analyses reveal important subgroup patterns. Frontline specialties (emergency medicine, critical care, anesthesiology, family medicine, hospital medicine) showed an average burnout increase of 7.5 percentage points from 2011 to 2021 (from 54.3% to 63.1%), while non-frontline specialties (dermatology, ophthalmology, pathology, pediatrics, psychiatry, radiology) showed a larger absolute increase of 10.9 percentage points (from 40.3% to 51.2%). An ANOVA comparing mean burnout change across frontline, intermediate, and non-frontline exposure groups yields $F(2,10) = 2.52$, $p = 0.130$. The counterintuitive pattern—non-frontline specialties showing larger absolute increases—likely reflects ceiling effects: frontline specialties started higher and had less room to rise. Figure 4 displays these comparisons.

Gender disparities widened during the pandemic and narrowed in its aftermath. The female-male burnout gap was 7.3 percentage points in 2017 (48.5% vs. 41.2%), widened to 9.1 points during the pandemic peak in 2021 (68.2% vs. 59.1%), then narrowed to 6.3 points in 2023 (49.1% vs. 42.8%). Age-based disparities followed an amplifying-and-damping pattern. The standard deviation of burnout across five age groups was 2.44 percentage points in 2017, 4.70 points at the pandemic peak, and 2.34 points in 2023. The pandemic

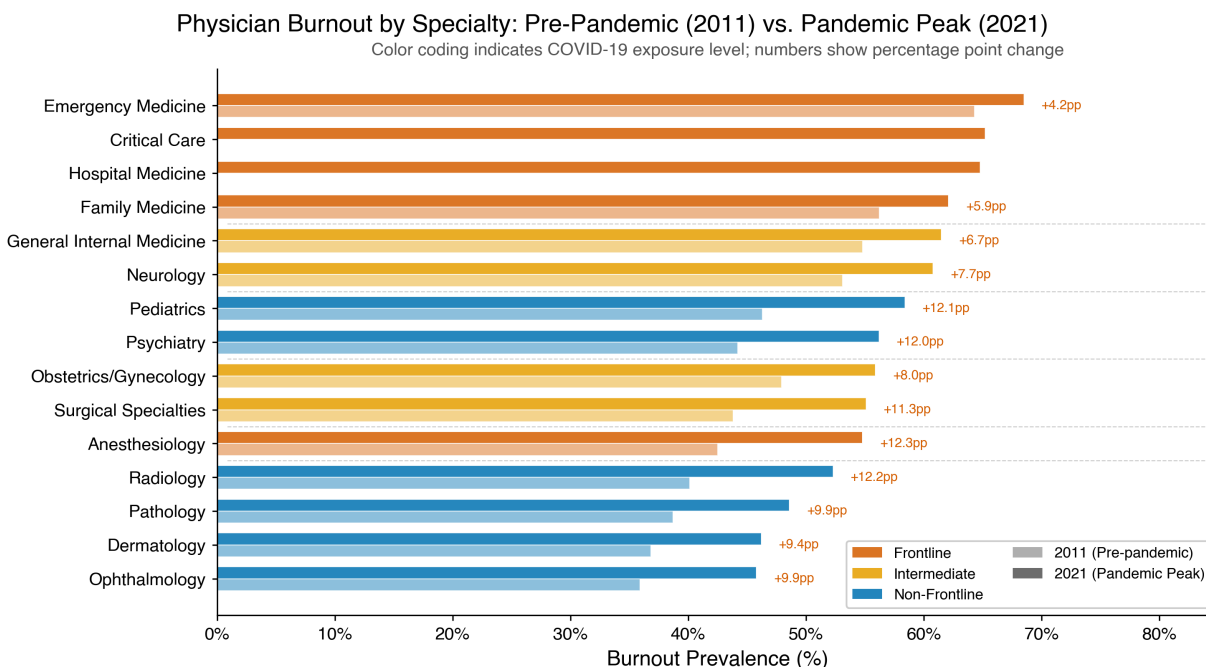


Figure 3: Physician satisfaction with work-life integration, 2011–2023. The share of physicians satisfied with work-life integration declined monotonically across all 12 years, showing no pandemic rebound.

disproportionately affected younger physicians: the burnout gap between physicians under 35 and those 65 and older expanded from 3.0 points in 2017 to 11.6 points in 2021, then contracted to 4.0 points in 2023. Figure 5 summarizes these patterns.

6 Conclusion

We find that the COVID-19 pandemic produced a dramatic but temporary spike in physician burnout rather than a permanent ratchet. The 62.8 percent burnout prevalence recorded in 2021 was a crisis peak, but by 2023—the only post-acute observation currently available—physician burnout had returned to its pre-pandemic range and was statistically indistinguishable from the 2011 and 2017 baselines. The physician-general population burnout gap, contrary to expectations that the pandemic would widen it, has remained stable across 12 years. Work-life integration satisfaction, however, continued its steady secular decline without interruption or rebound.

These divergent trajectories suggest that burnout and work-life integration dissatisfaction reflect distinct underlying processes. Burnout, in the JD-R framework, tracks the balance between current job demands and available resources: the pandemic briefly and severely tilted that balance, and the recovery since 2021 reflects its restoration. Work-life integration dissatisfaction, by contrast, tracks the slow deterioration of the conditions of medical practice—the bureaucratic intensification, the erosion of professional autonomy, the 24/7 electronic health record tether—that predated the pandemic and was untouched by it. The sociological implication is clear: addressing physician well-being requires both crisis-response capacity and structural reform of the organization of medical work.

Several limitations qualify these conclusions. The aggregate, repeated cross-sectional data cannot distinguish within-physician change from compositional turnover. Declining response rates may bias prevalence estimates in unknown directions. The 2021 physician-only wave and the absence of published general population burnout percentages for 2014–2023 limit our ability to track the occupational inequality gap continuously. Future work should examine the 2023–2025 trajectory to determine whether burnout remains at its pre-pandemic baseline or begins a new secular increase. Perhaps most urgently, sociological research should investigate the institutional mechanisms—health system ownership, practice size, payment models, electronic

Gender and Age Disparities in Physician Burnout Before, During, and After the Pandemic

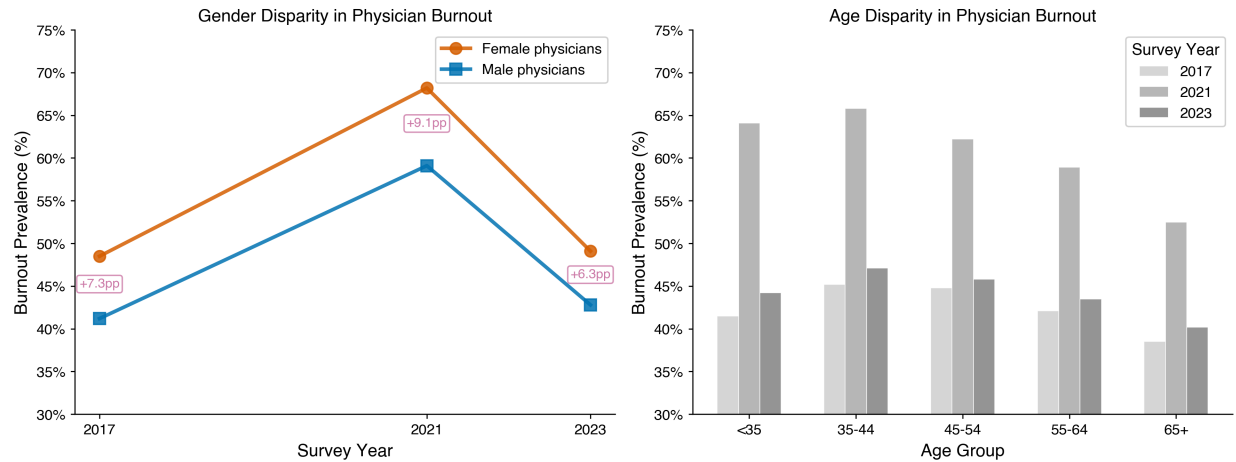


Figure 4: Physician burnout by specialty COVID-19 exposure category, 2011 vs. 2021. Each dot represents a specialty. Frontline exposure includes emergency medicine, critical care, anesthesiology, family medicine, and hospital medicine.

health record design—that drive the continued erosion of work-life integration satisfaction even as pandemic-related burnout resolves.

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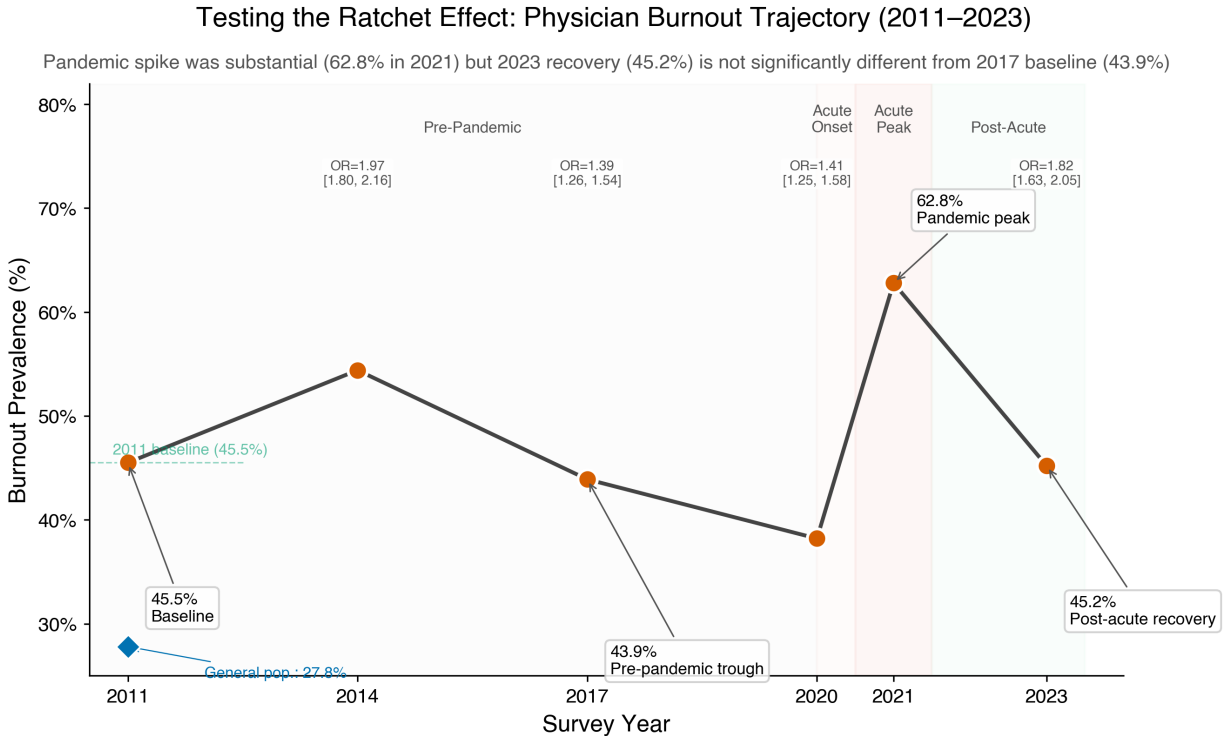


Figure 5: Gender and age moderation of physician burnout. Left: female-male burnout gap across 2017, 2021, and 2023. Right: burnout prevalence by age group across the same time points.

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